

A Packing of 11 Four-Spaces in $\mathbb{R}^{11}$

Limin Ge*

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Abstract

We give 11 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.799738. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.776562, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 11 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.799738 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 11$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.776562, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 11/10 = 2.800000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N11.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
279973844424470217295040690472987681158626245895453462974918981500982944
331164575881113098190296181910144456571071053694535104178032220306445439
9110798677956891247453414091213473564546116984212
```

denominator:

```
100000000000223982209289110614260359645288459066545479394202105147686608
596112565964472847907082468304524939839580973044897920715613081158884903
2709772847810274513807568032895842049472494782769
```

B Coordinates

11 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N11.txt`.

```
25692174855 -296598247766 -349889473035 -1820674690005298568407 -96682122678 -452109966035 -757110431154324461734 -246740889564 -126625593094 -200181036418
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-380340368551 -647533933818 126996399028 2463964200336380767844 573421486685 -132620329714 7524333506 192156421706 382612396473 63399997594 -406652893189
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```